Task-Relative Descriptive Privilege in Linear Gaussian Dynamical Systems

Abstract

In a finite-dimensional linear Gaussian dynamical system, the rank-r linear observer minimizing squared-error prediction loss is uniquely determined—up to rotation within the eigenspace—by the top-r eigenspace of the target information matrix. Two prediction targets with distinct dominant eigenspaces admit no jointly optimal observer. Global privilege—a single observer simultaneously optimal for all tasks—holds exactly when the target family is *r-coherent*: every information matrix shares a common top-r eigenspace. This is an exact algebraic condition. Numerical illustrations confirm a no-global-privilege instance ($\Gamma = 0.026 > 0$) and a structurally coherent construction, both consistent with the theorem.

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1. Introduction

A basic question in the study of dynamical systems is whether there is a single low-dimensional description that is optimal for every prediction task, or whether the optimal description depends on the target one seeks to predict. This paper answers that question exactly for finite-dimensional linear Gaussian dynamical systems (LGDS). The optimal rank-r observer is the top-r eigenspace of a target information matrix. Two targets whose information matrices have distinct dominant subspaces cannot be simultaneously optimised by any single observer. The exception has an exact algebraic characterisation: r-coherence of the target family.

The linear state-space framework is classical [1]; the relevant matrix optimisation on the Grassmannian [2] is governed by the Ky Fan maximum principle [3]. The novelty is not the Ky Fan principle or the optimality

of PCA for squared-error loss [4]—both classical—but the exact algebraic characterisation of when a single observer suffices across a target family: the r-coherence condition (Definition 3) and its biconditional (Corollary 4).

The result yields an exact risk formula, a no-global-privilege corollary, and an exact algebraic criterion for the exceptional coherent case.

This result is intentionally narrow. It concerns only finite-dimensional linear Gaussian dynamical systems, fixed-rank linear observers, and squared-error prediction targets.

2. System and Observer Family

Dynamics.. Let

$$x_{t+1} = Ax_t + \xi_t, \quad \xi_t \sim \mathcal{N}(0, Q), \quad x_t \in \mathbb{R}^n, \tag{1}$$

with A stable (spectral radius < 1) and $Q \succ 0$. The unique stationary covariance Σ satisfies $A\Sigma A^\top + Q = \Sigma$ and is strictly positive definite [1].

Observer family.. An observer of rank r ($1 \leq r < n$) is parametrised by a row-orthonormal matrix $U \in \mathbb{R}^{r\times n}$, $UU^\top = I_r$:

$$\phi_U(x) = U\Sigma^{-1/2}x. \tag{2}$$

Pre-whitening by $\Sigma^{-1/2}$ places all observers on a common isotropic footing; in the static case ($A = I$, $\tau = 0$) this reduces to principal component analysis [4]. Two observers differing only by an orthogonal rotation of their row spaces are equivalent; the observer space is the Grassmannian $\mathrm{Gr}(r, n)$, a smooth compact Riemannian manifold of dimension $r(n-r)$, with metric and geodesic structure from $O(n)$ [2].

Target family.. A prediction target is a pair (C, τ) with $C \in \mathbb{R}^{p\times n}$ and integer $\tau \geq 1$:

$$y_{C,\tau} = Cx_{t+\tau}. \tag{3}$$

Performance is total mean squared prediction error. A target family $\mathcal{T} = \{(C_k, \tau_k)\}$ is any finite collection.

3. Theorem

3.1. Exact Bayes risk formula

Given observation $z = U\Sigma^{-1/2}x_t$, the MMSE predictor of $Cx_{t+\tau}$ is linear by Gaussian conjugacy. At stationarity, $\mathrm{Cov}(Cx_{t+\tau},\, z^\top) = CA^\tau\Sigma^{1/2}U^\top$ and $\mathrm{Cov}(z, z^\top) = I_r$. The Bayes risk is

$$\mathcal{R}(U;\, C, \tau) = \mathrm{tr}(C\Sigma C^\top) - \mathrm{tr}\big(U\, M_{C,\tau}\, U^\top\big), \tag{4}$$

where the *target information matrix* is

$$M_{C,\tau} = \Sigma^{1/2}(A^\tau)^\top C^\top C A^\tau \Sigma^{1/2} \;\in\; \mathbb{R}^{n\times n}. \tag{5}$$

$M_{C,\tau}$ is symmetric positive semidefinite of rank at most $\min(p, n)$. The formula is exact.

3.2. Top-eigenspace optimality

Proposition 1. *For fixed rank r, the observer minimising $\mathcal{R}(U;\, C, \tau)$ over $\mathrm{Gr}(r, n)$ has row space equal to the top-r eigenspace of $M_{C,\tau}$. The minimum risk is*

$$\mathcal{R}^* = \mathrm{tr}(C\Sigma C^\top) - \sum_{i=1}^{r} \lambda_i(M_{C,\tau}), \tag{6}$$

where $\lambda_1 \geq \cdots \geq \lambda_n$ are the eigenvalues of $M_{C,\tau}$. Uniqueness up to rotation holds when $\lambda_r > \lambda_{r+1}$.

Proof. Minimising $\mathcal{R}$ is equivalent to maximising $\mathrm{tr}(UM_{C,\tau}U^\top)$ subject to $UU^\top = I_r$. By the Ky Fan variational principle [3, 5] the maximum equals $\sum_{i=1}^{r} \lambda_i(M_{C,\tau})$, achieved exactly when the row space of U spans the corresponding eigenspace. ■

3.3. No-global-privilege corollary

Corollary 2. *If targets (C_1, τ_1) and (C_2, τ_2) have information matrices M_1 and M_2 with distinct top-r eigenspaces, no single rank-r observer is simultaneously optimal for both.*

Proof. Optimality for target k requires the row space of U to equal the top-r eigenspace of M_k. Distinct eigenspaces cannot be spanned by the same U. ■

The corollary extends to any finite target family: if $\{M_k\}$ do not share a common top-r eigenspace, no rank-r observer is globally optimal.

3.4. r-coherence: the exact exception

Definition 3. *A target family $\mathcal{T}$ is* r-coherent *with respect to (A, Σ) if all matrices $\{M_{C,\tau} : (C, \tau) \in \mathcal{T}\}$ share a common top-r eigenspace V^*.*

Corollary 4. *Global privilege holds if and only if $\mathcal{T}$ is r-coherent.*

r-coherence requires all targets to be linear functionals of a common r-dimensional sufficient statistic across all horizons—a strong structural alignment between the dynamics and the task family. Outside r-coherence, the optimal observer generally depends on the target within the class studied here.

Remark 5. *r-coherence is an algebraic equality between eigenspaces, not an approximate or asymptotic condition. It is the exact characterisation of when global privilege is warranted.*

4. Numerical Illustrations

The theorem is analytically exact under its stated assumptions. We illustrate it numerically with two finite-dimensional instances: a no-global-privilege example and a structurally coherent example. All reported quantities are computed exactly from the analytic Bayes risk formula and numerical optimization on the Grassmannian.

Setup.. State dimension $n = 8$, observer rank $r = 2$. The model is a finite-dimensional linear Gaussian dynamical system [6]. $A = VDV^{-1}$ with eigenvalues drawn from $U[0.3, 0.85]$, random non-orthogonal V (seed 100), $Q = 0.1\, I_8$. Σ is computed from the discrete Lyapunov equation; $\Sigma \succ 0$ is confirmed to machine precision. All reported quantities are computed exactly from the analytic Bayes risk formula, eigenspace calculations, and repeated projected optimization over $\mathrm{Gr}(2, 8)$ (300 random initializations of projected L-BFGS-B, gradient-norm threshold 10^{-12}, function-value tolerance 10^{-15}).

Condition A: no-global-privilege instance.. Targets $T_1 = ([e_1^\top; e_2^\top], 1)$ and $T_2 = ([e_5^\top; e_6^\top], 5)$.

With non-normal A, A^1 and A^5 rotate state-space directions differently; the principal angles [7] between the top-2 eigenspaces of M_1 and M_2 are $54.1°$ and $24.0°$, confirming that no single rank-2 observer is simultaneously optimal.

Table 1: Regret matrix, Condition A ($n = 8$, $r = 2$, seed 100). All values from the exact analytic Bayes risk formula.

Observer	Regret on T_1	Regret on T_2
U_1^* (optimal for T_1)	0.000	0.036
U_2^* (optimal for T_2)	0.763	0.000
U_{joint} (compromise)	0.026	0.026

The global-privilege gap

$$\Gamma = \min_{U \in \mathrm{Gr}(r,n)} \max_k \left[\mathcal{R}(U;\, C_k, \tau_k) - \mathcal{R}_k^* \right] \tag{7}$$

measures the best achievable worst-case regret under task conflict. The minimax observer U_{joint} achieves $\Gamma = 0.026 > 0$. No globally optimal observer exists for this target family. The positive gap certifies the impossibility of any jointly optimal rank-2 observer for this task pair. Table 1 gives the full regret structure. The balanced regrets at U_{joint} (0.026 on both targets) are a necessary condition for a minimax with $\Gamma > 0$, confirming the global minimum was found. The large asymmetry in the off-diagonal entries (0.763 vs. 0.036) reflects the principal angle structure: the larger 54.1° principal angle shows U_2^*'s subspace is substantially misaligned with M_1's top eigenspace, forfeiting most of its eigenvalue weight, while the smaller 24.0° angle means U_1^* retains partial alignment with M_2's top eigenspace.

Condition B: structurally coherent example.. A target family is constructed so that all M_k share the same top-2 eigenspace by design: normal $A_{\text{sym}} = VDV^\top$ with orthogonal V, and a common C whose rows are the top-2 eigenvectors of A, varying only the prediction horizon $\tau \in \{1, 2, 3, 5\}$. Under this construction $M_{C,\tau}$ has the same dominant 2-dimensional eigenspace for all τ, so the four optimal observers are identical by the theorem. The maximum pairwise principal angle across all four is 0.000° to machine precision.

Under $\varepsilon = 0.1$ perturbations to C (20 random draws), the maximum principal angle across all draws and all pairs is 0.102 rad—well below the fragility threshold of 0.15 rad, set at 1.5ε. Structural coherence is perturbation-robust, distinguishing it from accidental near-coincidences of eigenspaces that break under small perturbations.

The numerical results are consistent with the theorem in all cases.

5. Scope

What is earned.. In finite-dimensional linear Gaussian dynamical systems at stationarity, among rank-r linear observers $\phi_U(x) = U\Sigma^{-1/2}x$ and squared-error prediction targets $y = Cx_{t+\tau}$, descriptive privilege is task-relative unless the target family is r-coherent. r-coherence requires strong structural alignment between dynamics and tasks. Outside it, no fixed-rank observer is globally optimal, and the cost of forcing any observer onto a non-native target is given exactly by the Bayes risk formula. The Grassmannian $\mathrm{Gr}(r, n)$ is the literal domain of this optimisation: a compact Riemannian manifold on which observer quality is a smooth function, with $O(n)$ acting by transformation. The computation reduces to a single $n \times n$ eigendecomposition of $M_{C,\tau}$, independent of τ once A^τ is formed; A^τ is obtained in $O(n^3 \log \tau)$ operations by repeated squaring.

What is not earned.. This result does not apply to nonlinear systems, non-Gaussian noise, nonlinear observers, or non-prediction tasks. The breakdown is structural: for non-Gaussian noise or nonlinear dynamics, the MMSE predictor is generally nonlinear and Proposition 1's trace maximisation has no closed-form counterpart; for nonlinear observers, the Grassmannian parametrisation and the Ky Fan principle both cease to apply. The causal axis—whether the observer optimal for intervention differs from the observer optimal for prediction—is not addressed here.

Relation to a broader programme.. This result establishes an exact tractable baseline. The linear Gaussian setting admits closed-form analysis because the MMSE predictor is linear, the stationary distribution is Gaussian, and the optimal observer reduces to a single eigendecomposition. Whether analogous structural conditions govern non-Gaussian or nonlinear systems—and whether the causal axis introduces further constraints on global privilege—are open questions motivating further empirical work.

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